wing lift curve slope: $a_w = \dfrac{2\pi AR}{2+\sqrt{4+AR^2}}$

Tail lift curve slope: $a_t = \dfrac{2\pi AR_{HT}}{2+\sqrt{4+AR_{HT}^2}}$

Downwash gradient: $\dfrac{d\epsilon}{d\alpha} = \dfrac{2a_w}{\pi AR}$

x_{ac} wing Aerodynamic center location: $x_{ac,w} = x_{leading\ edge} + 0.25\bar{c}$ of the MAC (LEMAC)

$y_{MAC} = \dfrac{b}{6} \cdot \dfrac{1+2\lambda}{1+\lambda}$, the spanwise location of where MAC is

$x_{LEMAC} = x_{root,LE} + y_{mac}\tan(\Lambda_{LE})$

- Location of neutral point based on entire aircraft

$x_{NP} = x_{ac,w} + \eta_t \cdot V_H \left(\dfrac{a_t}{a_w}\right)\left(1 - \dfrac{d\epsilon}{d\alpha}\right)\bar{c}$

- Static margin, distance between CG & NP

$SM = \dfrac{x_{NP} - x_{CG}}{\bar{c}}$

- AFT CG limit

$x_{CG,AFT} = x_{NP} - SM_{min} \cdot \bar{c}$

- Choose the SM_{min} in requirements that represent our aircraft

- Forward CG limit

$x_{CG,forward} = x_{ac,w} + \dfrac{L_t l_t + M_{ac,w}}{L_w}$

- L_w = wing lift
- L_t = tail force
- l_t = tail moment arm
- $M_{ac,w}$ = wing pitching moment $= C_{m,ac}\, q\, S_w \bar{c}$

$q = 0.5\rho v^2$, use website to find moment

- CG range $= x_{CG,AFT} - x_{CG,forward}$

- Performance Verification:

~~[illegible]~~

- cruise stall speed: find $C_{L,max}$ through website
- Find max velocity & do 83% to be cruise speed
- Rate of climb $= \dfrac{P_{available} - P_{required}}{W}$
  - P = thrust $\cdot$ velocity
- range $= \dfrac{\eta_p}{c}\left(\dfrac{L}{D}\right)\ln\left(\dfrac{W_i}{W_f}\right)$
- $\dfrac{\eta_p}{c} \cdot \dfrac{C_L^{3/2}}{C_D}\ln\left(\dfrac{W_i}{W_f}\right)$ = Endurance
- Takeoff distance $= \dfrac{1.44 \cdot F_G^2}{g \cdot \rho \cdot S \cdot T \cdot C_{L,max}}$
- Velocity TO $= \sqrt{\dfrac{2W}{\rho S C_{L,max}}}$
- Approach/Air distance from 50 ft obstacle, $S_A \approx \dfrac{height(obstacle)}{\tan(\gamma)}$
- γ = glide path angle